INTERNSHIP REPORT

A report submitted in partial fulfillment of the requirements for the Award of Degree of

BACHELOR OF ENGINEERING

in

CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Submitted by

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Under the supervision of

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(Duration: 06/10/2025 to 10/12/2025)

Learn Beyond

KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

AVINASHI ROAD, ARASUR

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2026

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BONAFIDE CERTIFICATE

This is to certify that the **Internship** report submitted by **DHARSHNA KUMAR K S (24AM012)** is work done by him and submitted during the academic year 2025 – 2026, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **CSE (Artificial Intelligence and Machine Learning)** at **Infosys Limited, Bengaluru.**

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Place: Coimbatore

Date: 27/02/2026

ACKNOWLEDGEMENT

First, I would like to thank **Mr. Satheesha B. N, Senior Vice President and Head - Education, Training and Assessment, Infosys Limited** for giving me the opportunity to undergo an internship within the organization.

I would also like to thank all the people who worked along with me in **Infosys Limited, Bengaluru** for their patience and openness in sharing knowledge they created for an enjoyable working environment. It is indeed with a great sense of pleasure and immense sense of gratitude that I acknowledge the help of these individuals.

I am highly indebted **Dr. K.P.Ramasamy**, Chairman, KPR Institutions, **Dr. A.M.Natarajan** Chief Executive, and **Dr. Devi Priya R,** Principal, KPR Institute of Engineering and technology for the facilities provided to accomplish the internship.

I would like to thank **Mr. Pandiya Rajan G,** Head of the Department, CSE (AIML) for his supportive actions by permitting me for my internship.

I would like to thank **Dr. Anandkumar R,** Associate Director, Industry Institute Partnership Cell (IIPC) for his support and advice to get internship and complete the same in above said organization. I am extremely grateful to **Mr. Anandhakrishnan T**, Department IIPC coordinator, faculty members and friends who helped me in the successful completion of this internship.

DHARSHNA KUMAR K S

ABSTRACT

The internship offers a dynamic and engaging experience in real-time software development, Artificial Intelligence integration, Python programming, and database management. The intern will actively participate in full-stack web development and gain hands-on experience working on an exciting Python project, with a specific focus on building a Hybrid AI Chatbot for banking workflows.

The internship helped me to understand and optimize backend server architectures. It also helped me learn about modern cloud databases and API integration used in the tech industry. One of the major milestones included developing a custom "Credit Saver" algorithm and a secure Admin Dashboard. This internship involved creating routing logic, managing session states, and ensuring an engaging and effective user interface for banking customers.

I had an opportunity to work on a real-world Python project, applying my coding skills to solve complex problems like API latency and cost optimization. This included collaborating and utilizing frameworks like Flask and Scikit-Learn to design, develop, and deploy a robust AI-based web application from scratch.

Generative AI is at the forefront of modern technology, and this internship offered a unique opportunity to explore Large Language Models (LLMs). I gained hands-on experience in machine learning intent detection, prompt engineering, and context injection, working on a project that leverages the capabilities of the Google Gemini API alongside local AI models.

Overall, this internship provided a multifaceted learning experience, combining Backend Architecture, Python software development, and cutting-edge Generative AI technology, setting the stage for a successful career in Full-Stack Engineering and Artificial Intelligence.

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Chapter 1

About the Company

Infosys Springboard is a repository of world-class corporate knowledge and technical wisdom contributed by the best minds in the IT industry. Operating on a massive scale to bridge the digital skills gap, Infosys Springboard offers masterclasses, certification courses, project-based learning, and comprehensive Virtual Internship programs (like Internship 6.0).

Overall, Infosys Springboard provides a host of learning modules for Students, Graduates, Academic Institutions, and emerging professionals. The courses and internship problem statements are custom- curated by industry experts and Infosys thought leaders to align perfectly with current corporate standards and real-world business requirements.

The Infosys Springboard Virtual Internship 6.0 program is actively involved in providing hands-on corporate exposure and development experience in emerging tech fields such as Artificial Intelligence, Natural Language Processing, Full-Stack Development, and Data Science.

Chapter 2

Plan for Internship

My 8-week virtual internship program at Infosys Springboard offered a holistic experience in full-stack web development, machine learning, and Generative AI integration. In the initial two weeks, I underwent a comprehensive orientation and requirement analysis phase, acquainting myself with the project scope for "BankBot" and setting up the core development environment. This included gaining preliminary exposure to relational database design using PostgreSQL and establishing the backend server fundamentals with Python and the Flask framework.

The following four weeks focused heavily on core AI and backend development, during which I conceptualized and built the "Hybrid Brain" architecture for the chatbot. This period helped me learn the intricacies of building a local machine learning intent-detection model using Scikit-Learn. Subsequently, I transitioned to cloud AI integration, working to seamlessly connect the Google Gemini API to our backend to tackle real-world banking queries, write context-aware prompts, and manage session states. This hands-on experience was designed to sharpen my coding, system architecture, and problem-solving skills.

The internship's final two weeks were all about frontend integration, optimization, and system deployment. Engaging in intensive testing and debugging, I developed a responsive chat interface and a secure admin dashboard, alongside implementing cost-saving algorithms like the "Credit Saver" logic. Throughout the internship, regular milestone submissions and evaluations ensured continuous professional growth. By the end of the internship, I gained a multifaceted skill set and a deep practical understanding of Python programming, database management, and hybrid Artificial Intelligence technologies.

<u>Weekly Overview of Internship Activities</u>

S. No.	Date	Name of the Module(s) Completed
Week 1 < 06/10/2025 – 11/10/2025>		
1.	06/10/2025	Program Orientation and Project Kickoff
2.	07/10/2025	Problem Statement Analysis: Hybrid AI Chatbot ("BankBot")
3.	08/10/2025	Setup of Python Development Environment and IDE (VS Code)

4.	09/10/2025	Version Control Initialization (Git & GitHub Setup)
5.	10/10/2025	Foundational Research: LLMs, Gemini API, and Machine Learning
6.	11/10/2025	Week 1 Mentor Catch-up & System Architecture Design
Week 2 < 13/10/2025 – 18/10/2025>		
7.	13/10/2025	Database Selection: Setting up PostgreSQL on Neon Cloud
8.	14/10/2025	Designing Relational Database Schema for Banking Data
9.	15/10/2025	Creating Core Tables (customers, branches, logs)
10.	16/10/2025	Implementing Python Database Connectivity using psycopg2
11.	17/10/2025	Populating the Database with Mock Banking Test Data
12.	18/10/2025	Week 2 Review & Database Module Finalization
Week 3 < 20/10/2025 – 25/10/2025>		
13.	20/10/2025	Backend Framework Initialization (Python Flask Setup)
14.	21/10/2025	Building Core Server Application (app.py backbone)
15.	22/10/2025	Creating Backend API Routing (/chat and /admin endpoints)
16.	23/10/2025	Testing Server Client-Server JSON Responses via Postman
17.	24/10/2025	Structuring Project Directories and Error Handling Modules
18.	25/10/2025	Week 3 Review & Basic API Functionality Testing
Week 4 < 27/10/2025 – 01/11/2025>		
19.	27/10/2025	Week 3 Review & Basic API Functionality Testing
20.	28/10/2025	Collecting and Formatting Training Data for Intent Detection
21.	29/10/2025	Developing the "Local Brain" Intent Classifier (train_brain.py)
22.	30/10/2025	Training and Serializing the Naive Bayes Model (local_brain.pkl)
23.	31/10/2025	Integrating the Local ML Model with Flask Chat Routing
24.	01/11/2025	Week 4 Review & Local Intent Detection Testing
Week 5 < 03/11/2025 – 10/12/2025>		
25.	03/11/2025 – 08/11/2025	Frontend UI, Cloud AI Integration & Final Deployment Phase
26.	10/11/2025 – 15/11/2025	Frontend Development: Building "Nova Finance" Login & Chat UI with Voice Input
27.	17/11/2025 – 22/11/2025	Admin Dashboard Implementation: Managing Branches, Customers, and Chat Logs
28.	24/11/2025 – 29/11/2025	System Integration: Connecting Flask Backend with Frontend UIs and "Credit Saver" Logic

29.	01/12/2025 – 06/12/2025	Comprehensive Testing, Debugging, and UI Polishing for Final Deliverable
30.	08/12/2025 – 10/12/2025	Project Documentation, Final Report Preparation, and Final Submission

Chapter 3

Training Task

As an intern, the eight-week journey ahead was a thrilling prospect for personal and professional development. The training tasks were thoughtfully designed to provide me with a well-rounded experience in full-stack web development, Python programming, and Artificial Intelligence integration. In the initial phase, the focus was firmly on backend architecture and database systems. Over the first three weeks, I immersed myself in the learning of Python and the Flask framework, gaining a deep understanding of backend routing, REST APIs, and their role in modern software. My tasks included setting up a PostgreSQL database on Neon Cloud, designing the relational schema for banking data (such as customers and branches), and establishing secure server connectivity. These activities equipped me with the essential skills needed to create a robust, scalable foundation for a complex application. The following two weeks pivoted toward Artificial Intelligence and machine learning. This phase excited me the most, as I received comprehensive training in building predictive models, providing a solid foundation for the coding challenges ahead. Engaging in hands-on coding practice, I developed the "Local Brain" intent-detection system using Scikit-Learn, and subsequently integrated the Google Gemini API to act as the "Cloud Brain". These tasks not only enhanced my programming skills but also offered a brilliant glimpse into the practical application of technology in solving complex business problems like API latency and cost optimization. In the final three weeks, the program took a fascinating turn as I focused on frontend integration, system optimization, and final deployment. Building a responsive Chat Interface and a secure Admin Dashboard was a unique hands-on experience. I learned about asynchronous JavaScript messaging, implemented the custom "Credit Saver" logic to bypass unnecessary cloud costs, and successfully tested the entire "Vaulty" banking ecosystem. This phase truly put my skills to the test, and the learning and growth it brought were invaluable.

Implementation

The implementation of the "Vaulty" AI Chatbot was systematically executed across four integrated modules: database architecture, local machine learning, cloud generative AI, and backend routing. First, a robust relational database was established using PostgreSQL to securely manage banking data and comprehensively log chat interactions. Next, to optimize API costs and reduce response latency for routine banking queries, a "Local Brain" was developed using Scikit-Learn. This local intent-detection model leveraged a Naive Bayes classifier to instantly recognize common customer requests, such as checking branch timings. For complex or highly specific banking inquiries that fell outside routine tasks, the system integrated the Google Gemini API as its "Cloud Brain," utilizing strict prompt engineering to ensure professional and accurate AI behavior. Tying these components together, the core application was built using the Python Flask framework. The Flask backend acted as an intelligent router, employing a custom "Credit Saver" algorithm to evaluate the user's input, check the local model's confidence score, and dynamically route the query to either the zero-cost local database or the advanced Gemini cloud model, thereby delivering a highly efficient, full-stack hybrid AI solution.

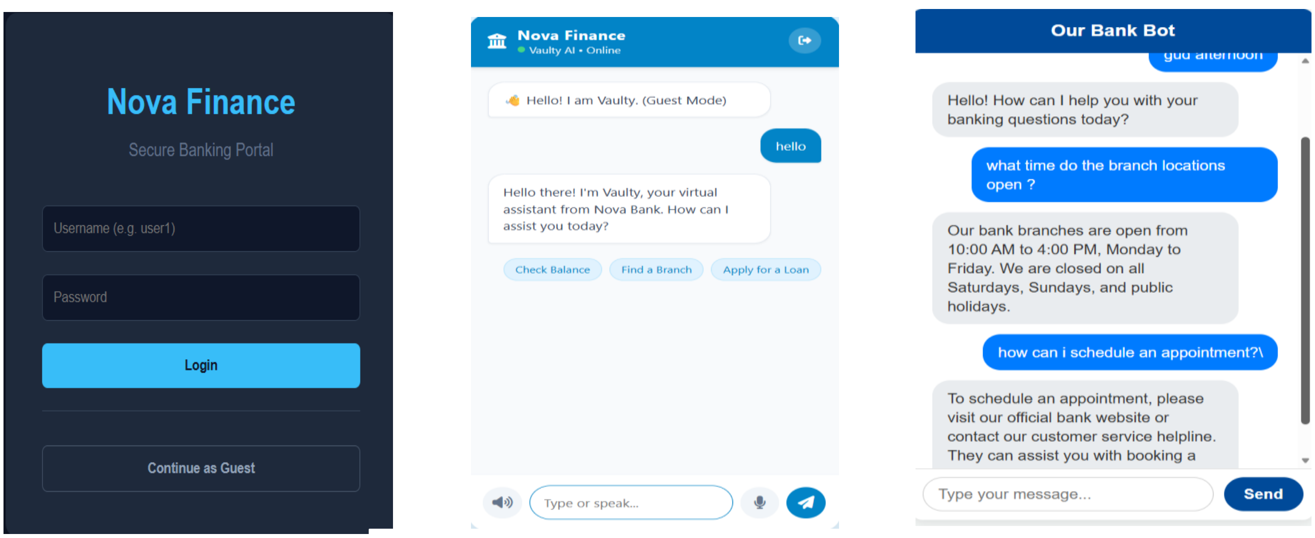


Fig 1 User Interfaces of Virtual Assistant

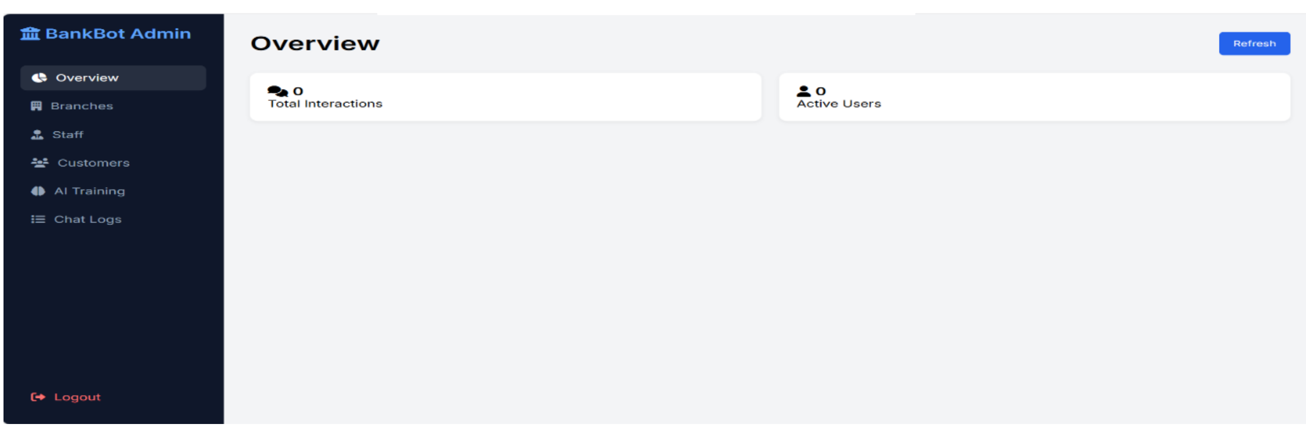


Fig 2 Admin Dashboard

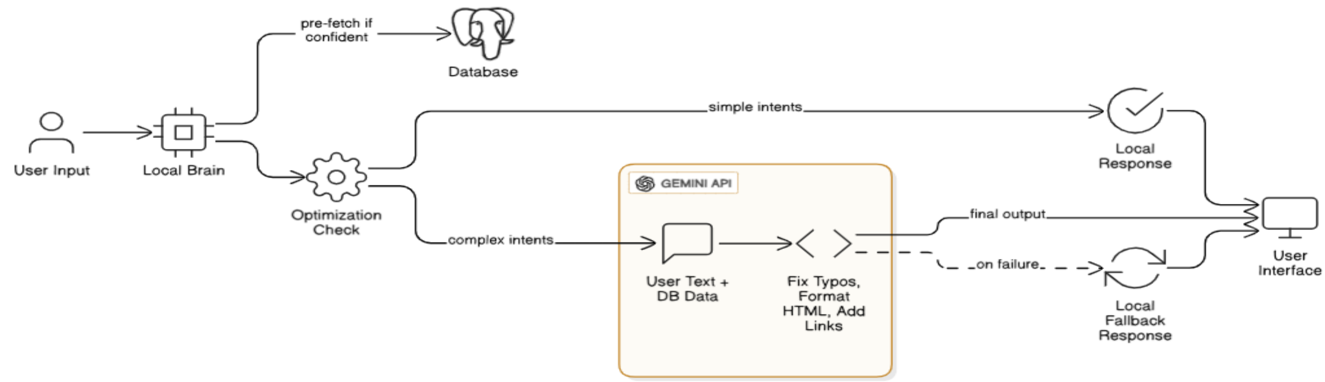


Fig 3 System Architecture

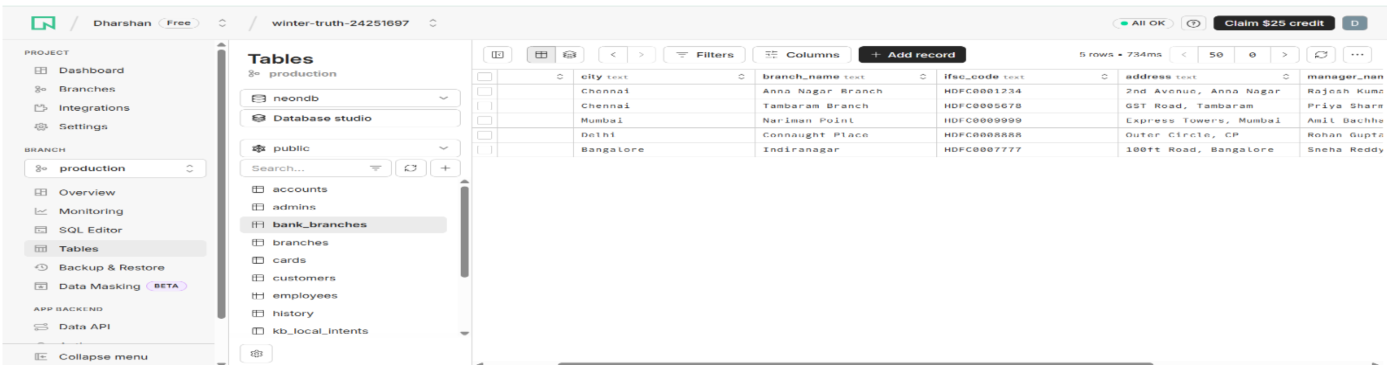


Fig 4 Database

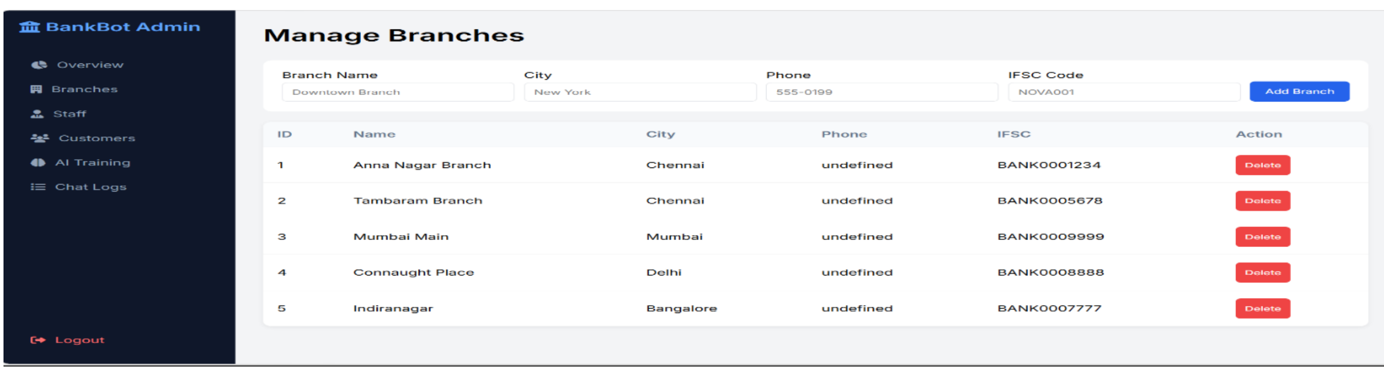


Fig 5 Admin Dashboard to manage branches details

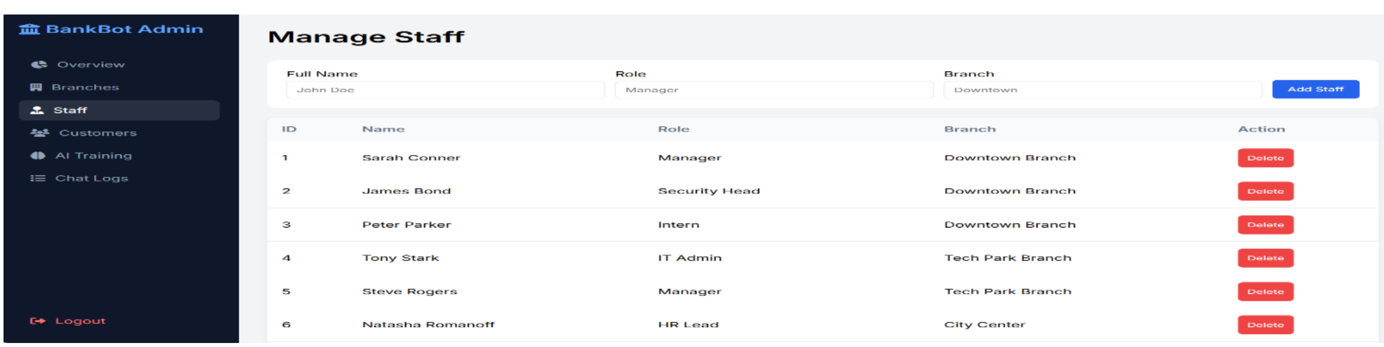


Fig 6 Admin Dashboard to manage Staff details

1. The Certificate & Completion Email

- The Email: This is your official completion notice from Infosys Springboard. It confirms your successful completion of the internship and specifies your project track: "BankBot: Development of an AI Chatbot for Banking FAQs". It also includes the strict instruction not to share project artifacts (like the code or the UI screenshots) on social media.
- The Certificate: This is the official document awarded to you, Dharshna Kumar K S, verifying your participation from October 6, 2025, to December 10, 2025. It is signed by the Head of Education, Training and Assessment at Infosys.

2. The Project User Interfaces (The "Vaulty" Application)

The final image showcases the actual front-end application you developed for your project. It is broken down into four distinct screens, which beautifully illustrate the "Full-Stack" nature of your work:

- Top Left (Login Portal): * This is the secure entry point for the "Nova Finance" banking application. It features standard authentication fields (Username/Password) and includes a "Continue as Guest" option, allowing users to access the chatbot without logging in.
- Top Middle (Vaulty Chat Interface - Welcome Screen):
  - This shows the chat interface when a user enters as a guest. Your bot introduces itself as "Vaulty."
  - It features quick-action buttons (Check Balance, Find a Branch, Apply for a Loan) for easy navigation.
  - Notice the microphone icon at the bottom, indicating you incorporated speech-to- text or voice input capabilities into the UI.
- Top Right (Active Chat Conversation):
  - This screen demonstrates the chatbot actively handling Banking FAQs (which perfectly matches your project title).
  - The bot successfully answers routine queries, such as branch opening hours (10:00 AM to 4:00 PM) and how to schedule an appointment. This is an excellent visual representation of the "Local Brain" intent-detection we discussed earlier in your report, handling standard FAQs instantly.
- Bottom (BankBot Admin Dashboard):
  - This is the secure backend panel for bank staff.
  - The sidebar menu reveals the comprehensive nature of your database

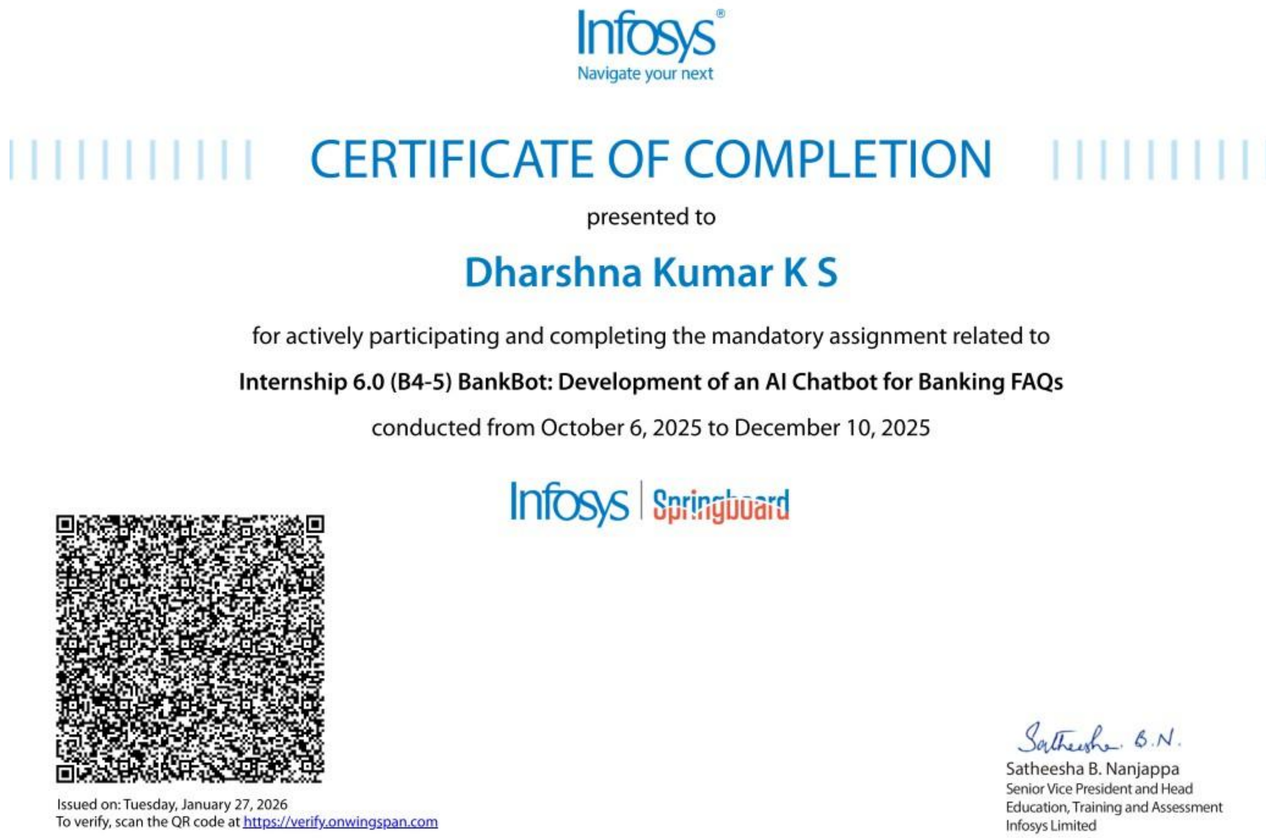
work, with options to manage Branches, Staff, Customers, AI Training, and Chat Logs.

- The main overview area is designed to track metrics like "Total Interactions" and "Active Users.

Chapter 4
Conclusion

In the beginning, the orientation and requirement analysis phase introduced me to the project scope and the essential concepts of full-stack web development and Artificial Intelligence. It was the perfect foundation on which to build my knowledge of Python backend architecture and database management. The subsequent weeks were all about hands-on learning. I had an opportunity to conceptualize a hybrid AI system, design a relational PostgreSQL database, and build a robust backend server using the Flask framework. These experiences not only improved my coding skills but also taught me how to approach and solve complex software engineering challenges, such as API latency and cost optimization, effectively. The integration of the Google Gemini API and local Machine Learning models was a fascinating eye-opener. I explored cutting-edge generative AI technology, delving into prompt engineering and implementing a custom "Credit Saver" algorithm. The "Vaulty" project helped me to showcase my skills in a practical application that merged secure database integration, Python backend development, and advanced AI concepts. Overall, this Infosys Springboard internship has been an invaluable experience. I leave with newfound knowledge and confidence, ready to tackle the professional tech world. I am grateful for the guidance and mentorship I received and excited to explore further opportunities in the fields of Full-Stack Development, Machine Learning, and Generative AI.

Internship Certificate



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presented to

Dharshna Kumar K S

for actively participating and completing the mandatory assignment related to

Internship 6.0 (B4-5) BankBot: Development of an AI Chatbot for Banking FAQs

conducted from October 6, 2025 to December 10, 2025

Infosys | Springboard

Issued on: Tuesday, January 27, 2026
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Satheesha B. Nanjappa
Senior Vice President and Head
Education, Training and Assessment
Infosys Limited